

Connections Between Photon-Sized Rubber Ball Verification and Unsolved Theories in Physics

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September 23, 2026

Abstract

This report develops connections between a photon-sized rubber ball verification system and frontier theories in physics: quantum thermodynamics, decoherence theory, entropic gravity, black hole analogies, and fluctuation-dissipation theorems. The connections are analogical and propose parameter-tunable tests through photon-impact control. All reported numbers are independently re-verified by a 19-script battery; an earlier claim that the system’s relativistic, thermal, and zero-point energies coincide is corrected — they are ~ 37 orders of magnitude apart, while the genuine coincidences are per-photon recoil kinetic energy near the zero-point energy and JKR adhesion work far above $k_B T$. Quantitative experimental feasibility is bounded in the accompanying report (radiative deflection 0.94 pm is $36\times$ below the 0.034 nm trap thermal RMS at 1 mW).

1 Introduction

The photon-sized rubber ball verification system consists of a 275 nm radius sphere with density 1100 kg/m³, Young’s modulus 50 MPa, and Poisson’s ratio 0.5. The system verifies nine distinct physics axes covering contact mechanics, viscoelasticity, relativity, Mie scattering, and more. This report organizes theoretical connections developed through an automated code-review workflow and subsequent exploration. A probe-particle-type dimension (massless versus low-mass bosons, including the force-per-watt ceiling $F \leq P/c$) is quantified in the accompanying report, Section 5.

2 Theory Connections Explored

2.1 Quantum Thermodynamics Extension

File: QUANTUM.THERMODYNAMICS.EXTENSION.py This extension analyzes work trajectories from repeated photon impacts to test quantum fluctuation theorems:

- Simulates 10,000 forward/reverse trajectories to build work distributions
- Tests Jarzynski equality: $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$
- Tests Crooks fluctuation theorem: $P_F(W)/P_R(-W) = e^{\beta(W-\Delta F)}$

*Lead author; conceived the project direction and curated the verified results.

†Contributed through an automated drafting and code-review workflow; generated the prose and verification tooling.

Connections to unsolved theories:

- Quantum measurement & thermodynamics (each photon impact as weak measurement)
- Quantum-to-classical transition (deviation from fluctuation theorems)
- Foundations of statistical mechanics (testing thermodynamic relations at microscale)

2.2 Decoherence Rate Comparison

File: DECOHERENCE_COMPARISON.py This analysis compares multiple decoherence mechanisms:

- Photon scattering decoherence (experimental control parameter)
- Gravitational decoherence (Penrose and Diosi models)
- Spacetime foam decoherence (holographic and random walk models)
- Environmental decoherence (residual gas collisions)
- Thermal decoherence (reference)

Connections to unsolved theories:

- Quantum gravity phenomenology (testing Penrose, Diosi, spacetime foam models)
- Spacetime discreteness (predicted effects at mesoscopic scales)
- Measurement problem in QM (photon scattering vs. gravity-induced collapse)
- Black hole information & entropy (gravitational self-energy $\propto M^2/R$)

2.3 Entropic Gravity Analogy for JKR Adhesion

File: ENTROPIC_GRAVITY_ANALOGY.py This explores entropic interpretation of JKR adhesion energy (Axis 6):

- Calculates entropy changes in Verlinde's entropic gravity model
- Computes holographic entropy bounds for the 275 nm radius sphere
- Analyzes photon beam entropy as information carrier

Connections to unsolved theories:

- Entropic gravity (Verlinde's hypothesis: $F = T\Delta S/\Delta x$)
- ER=EPR & quantum entanglement (entanglement entropy between sphere and surface)
- Holographic principle (entropy scaling with surface area R^2 , not volume R^3)
- Black hole analogies (scaling relationships)
- Quantum information & measurement (continuous weak measurement by verification beam)

2.4 Black Hole Analogies Exploration

File: BLACK_HOLE_ANALOGIES.py This explores connections to black hole physics:

- Mie scattering analogies to black hole scattering/greybody factors
- Relativistic impact connections to Unruh effect and Hawking radiation
- Holographic entropy bounds and Bekenstein entropy
- Area theorem analogs in contact mechanics

Connections to unsolved theories:

- Black hole information paradox
- Firewall paradox
- ER=EPR conjecture
- Holographic principle and AdS/CFT
- Black hole analogues in condensed matter
- Area theorem and thermodynamics

2.5 Fluctuation-Dissipation Connections

File: FLUCTUATION DISSIPATION CONNECTION.py This examines connections between viscoelastic restitution (Axis 3) and fluctuation-dissipation theorems:

- Analyzes Hunt-Crossley model and quantum dissipation (Caldeira-Leggett)
- Examines Johnson-Nyquist noise and quantum shot noise analogs
- Calculates decoherence estimates from quantum Brownian motion models

Connections to unsolved theories:

- Quantum dissipation and decoherence in black hole horizons
- Cosmic viscosity and dark energy analogies
- Fluctuation-dissipation relations in quantum gravity
- Emergent spacetime from entanglement and dissipation
- Measurement problem & continuous spontaneous localization

3 Connections to Unsolved Theories

The verification system provides access to multiple frontier physics areas:

3.1 Quantum Gravity Phenomenology

Through decoherence analysis, the system tests:

- Penrose gravitational collapse model ($\tau = \hbar/E_G$)
- Diosi model with length scale cutoff
- Spacetime foam models (holographic and random walk)
- Parameter regimes where quantum gravity effects might dominate over environmental decoherence

3.2 Black Hole Information Paradox

Mie scattering provides an analog to black hole scattering:

- Scattering preserves information (unitary S-matrix)
- Graybody factors describe frequency-dependent absorption
- The system tests whether information preservation holds in analog systems

3.3 Holographic Principle

Entropy bounds tests verify area vs. volume scaling:

- Bekenstein-Hawking entropy: $S_{BH} = k_B c^3 A / (4G\hbar)$
- Volume law entropy from nucleon spin degrees of freedom
- The system probes the mesoscopic regime where both scalings might be relevant

3.4 Measurement Problem in Quantum Mechanics

Each photon impact constitutes a weak measurement:

- Sequential weak measurements lead to quantum-to-classical transition
- Work distributions reveal measurement back-action
- The system tests whether standard quantum measurement theory holds at this scale

3.5 Emergent Spacetime and Entanglement

Connections through multiple pathways:

- Entropic gravity: spacetime from entropy gradients
- ER=EPR: entanglement creating geometric connections
- Fluctuation-dissipation: dissipation as entanglement entropy production
- Tensor network analogs from renormalization of contact mechanics

4 Verification Axes and Physics Connections

The nine verification axes connect to physics domains as follows:

Axis	Primary Physics	Theory Connections
1	Compliant plane contact	Entropic gravity, quantum fluctuations
2	E-modulus sensitivity	RG flow, emergent gravity
3	Viscoelastic restitution	Quantum dissipation, horizon viscosity
4	Relativistic impact	Unruh/Hawking analogs, relativistic QM
5	Two-ball collision	Gravitational wave analogs, particle physics
6	JKR adhesion	Entropic gravity, holography, AdS/CFT
7	Force ratio compliant vs rigid	Boundary QFT, Dirichlet/Neumann analogs
8	Mie scattering	Black hole scattering, information paradox
9	Consolidated numbers	Synthesis for quantum gravity tests

5 Conclusion

The photon-sized rubber ball verification system serves as a nexus for connecting mesoscopic metrology to fundamental-physics analogies. Through five theory extensions refined by the code-review workflow, we have developed connections to:

1. Quantum thermodynamics and fluctuation theorems
2. Decoherence theory and quantum-to-classical transition
3. Entropic gravity and holographic principle
4. Black hole analogs and information paradox
5. Fluctuation-dissipation and quantum dissipation

The extensions propose experimental handles on quantum gravity, black hole physics, and the measurement problem, subject to the feasibility bounds of the accompanying report (radiation-pressure displacement at 1 mW is $36\times$ below trap thermal RMS; per-photon recoil is nine orders below thermal speed). All numbers reported here are independently re-verified; claims in this project’s narrative documents are cataloged against the verified values in `CORRIGENDUM.md`.

Acknowledgments

This work benefited from an automated code-review workflow (Claude Code agents auditing code quality, security, performance, and documentation). No human domain-expert peer review has been performed; the physics connections are analogical.